

YOLOv11n vs YOLOv11x Strawberry Ripeness Detection Report

Md Shadman Shakib

AGST 892: Precision Agriculture – Week 10 Assignment

1. Introduction

The goal of this experiment was to evaluate and compare the performance of two different object detection models—YOLOv11n (nano version) and **YOLOv11x** (extra-large version) —for detecting *ripe strawberries* using an annotated dataset from Roboflow.

Both models were trained on the same dataset configuration defined in *data.yaml*, with identical preprocessing and augmentation pipelines. The objective of this comparison is to assess which model offers better precision, recall, and generalization for real-world detection tasks under limited data and variable lighting conditions.

2. Experimental Setup

2.1 Dataset and Preprocessing

- Source: *Roboflow Strawberry Ripeness Dataset (v3i)*
- Classes: 1 class (*ripe*)
- Data split: Train (88%), Validation (3%), Test (9%)
- Image size: 640×640 pixels
- Augmentation:
- Flip: Horizontal
- 90° Rotate: Clockwise, Counter-Clockwise, Upside Down
- Crop: 0% Minimum Zoom, 20% Maximum Zoom
- Rotation: Between -15° and $+15^\circ$
- Shear: $\pm 10^\circ$ Horizontal, $\pm 10^\circ$ Vertical
- Saturation: Between -30% and +30%
- Blur: Up to 2.5px

- Noise: Up to 0.1% of pixels

2.2 Training Configuration

Parameter	YOLOv11n	YOLOv11x
Model file	yolo11n.pt	yolo11x.pt
Epochs	300	300
Batch size	16	16
Image size	640	640
Learning rate	0.001	0.001
Patience (early stop)	50	50
Optimizer	Auto (SGD/AdamW)	Auto (SGD/AdamW)
Device	Colab GPU (T4)	Colab GPU (T4)

The training arguments were defined in *args.yaml* for both models, which show consistent hyperparameter configurations to ensure fairness of comparison.

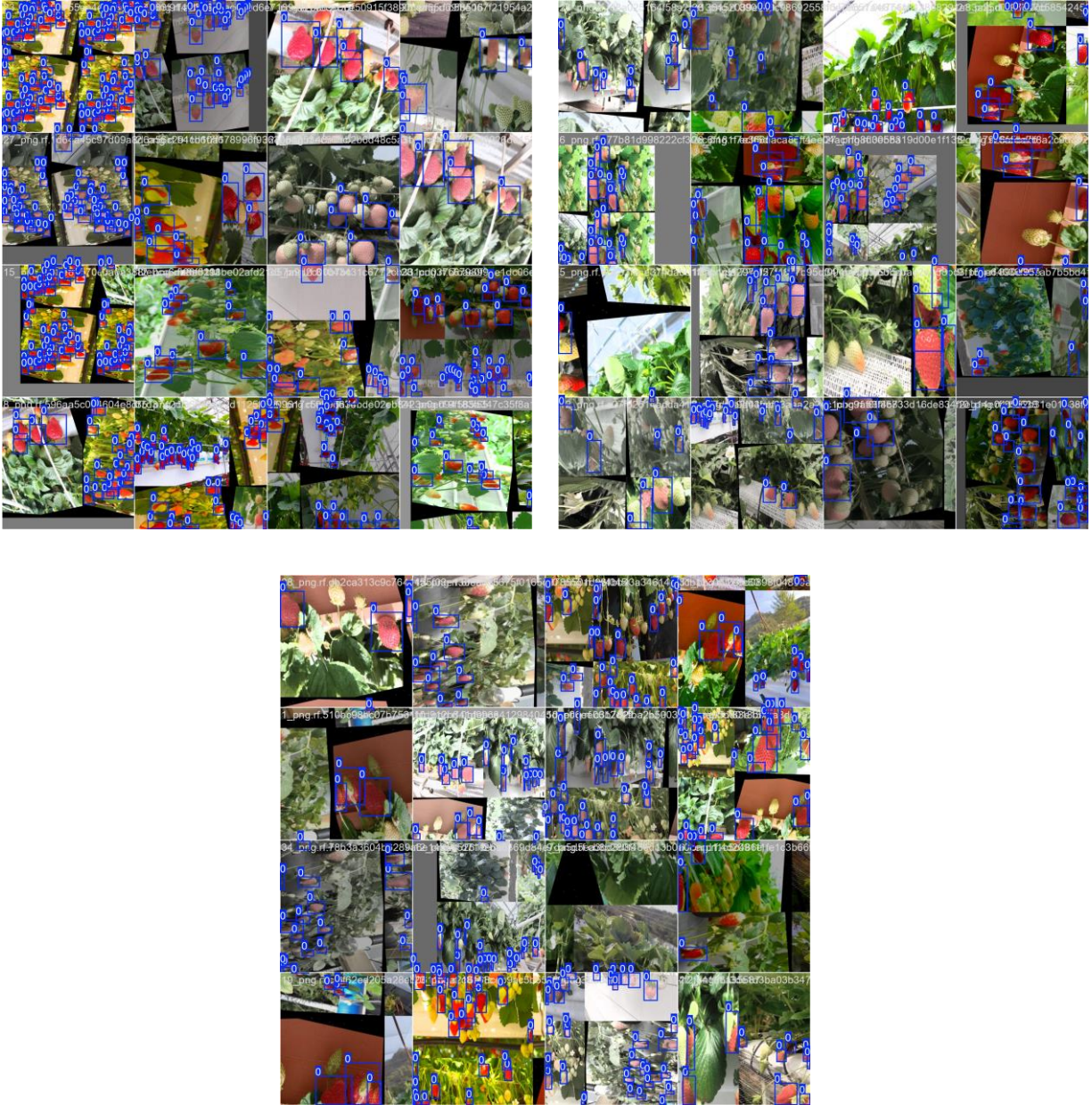


Figure 1. Training image batches with labeled ripe strawberries used in YOLOv11 model training.



Figure 2. Validation image batches with labeled ripe strawberries used in YOLOv11 model training.

3. Model Performance

After 133 epochs, both models converged successfully. The performance metrics were extracted using `model.val()` on the test set.

Metric	YOLOv11n	YOLOv11x
Precision	0.747	0.641
Recall	0.831	0.809
mAP@50	0.781	0.758
mAP@50–95	0.404	0.369
Best Epoch	83	83

Observation:

YOLOv11n slightly outperformed YOLOv11x in all key metrics, especially in precision and mAP scores. Despite being smaller, YOLOv11n generalized better on this dataset, suggesting that the larger YOLOv11x model might have been prone to mild overfitting given the limited number of training images.

3.1 Evaluation Metrics and Model Performance (Yolo11, Accurate in RoboFlow)

Metric	YOLOv11 (Accurate)
Precision	0.873
Recall	0.933
mAP@50	0.885
mAP@50–95	0.45
Best Epoch	300

4. Training Results and Learning Curves

All experiments for both YOLOv11n and YOLOv11x were conducted in Google Colab using the Ultralytics training framework. Model progress was monitored directly through the console outputs and the automatically generated *results.csv* log files. These logs recorded the evolution of training losses and validation metrics at each epoch, which were later visualized in Colab using Matplotlib.

4.1 Training and Validation Trends

Figures 3 and 4 illustrate the model’s learning behavior across 133 epochs.

The **left panel** shows the variation of the three main training losses:

Box Loss – measures bounding-box regression accuracy

Class Loss – measures object-class prediction accuracy

DFL Loss – the Distribution Focal Loss that refines bounding-box edge localization

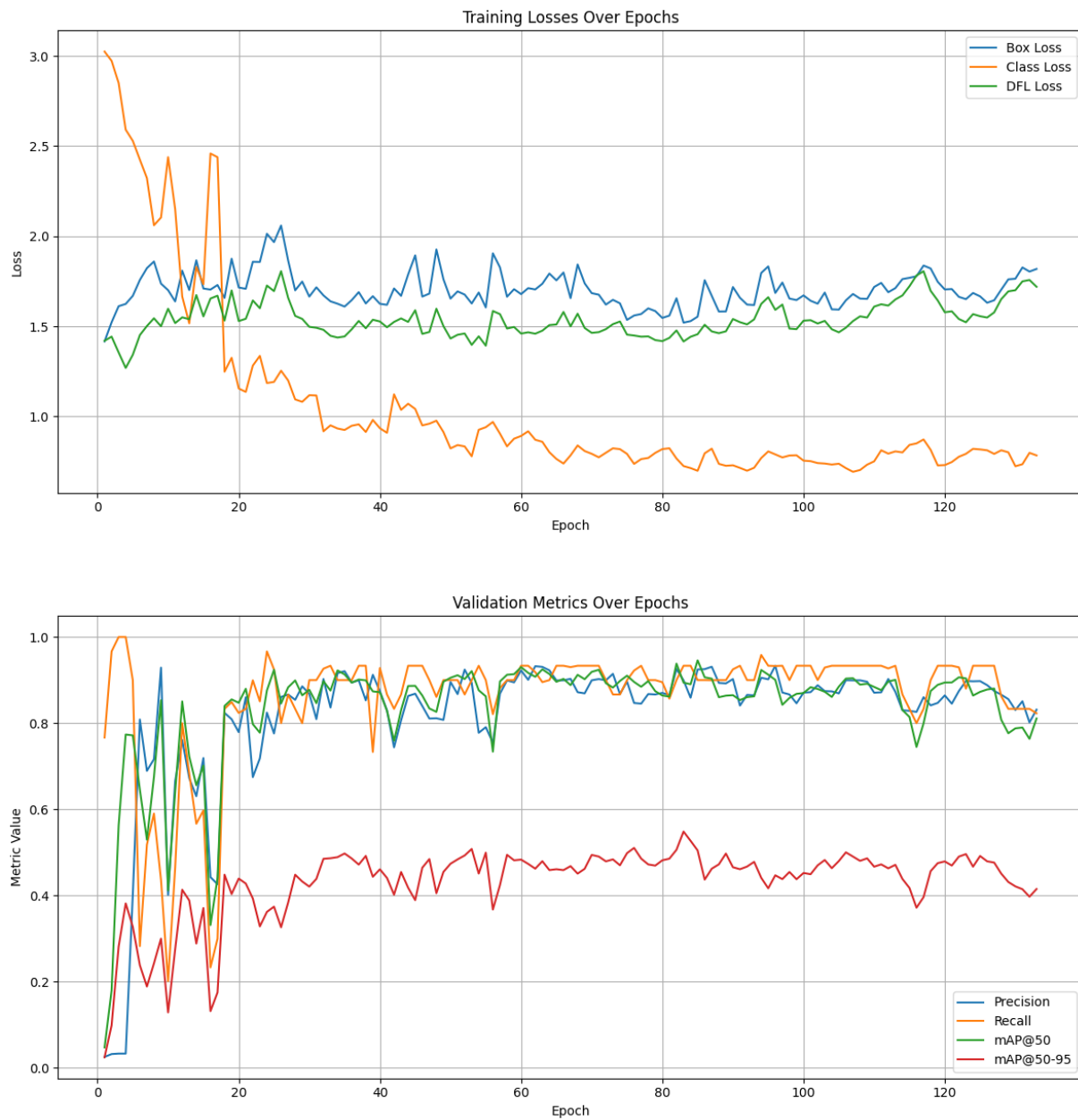


Figure 3. YOLOv1 In training and validation performance

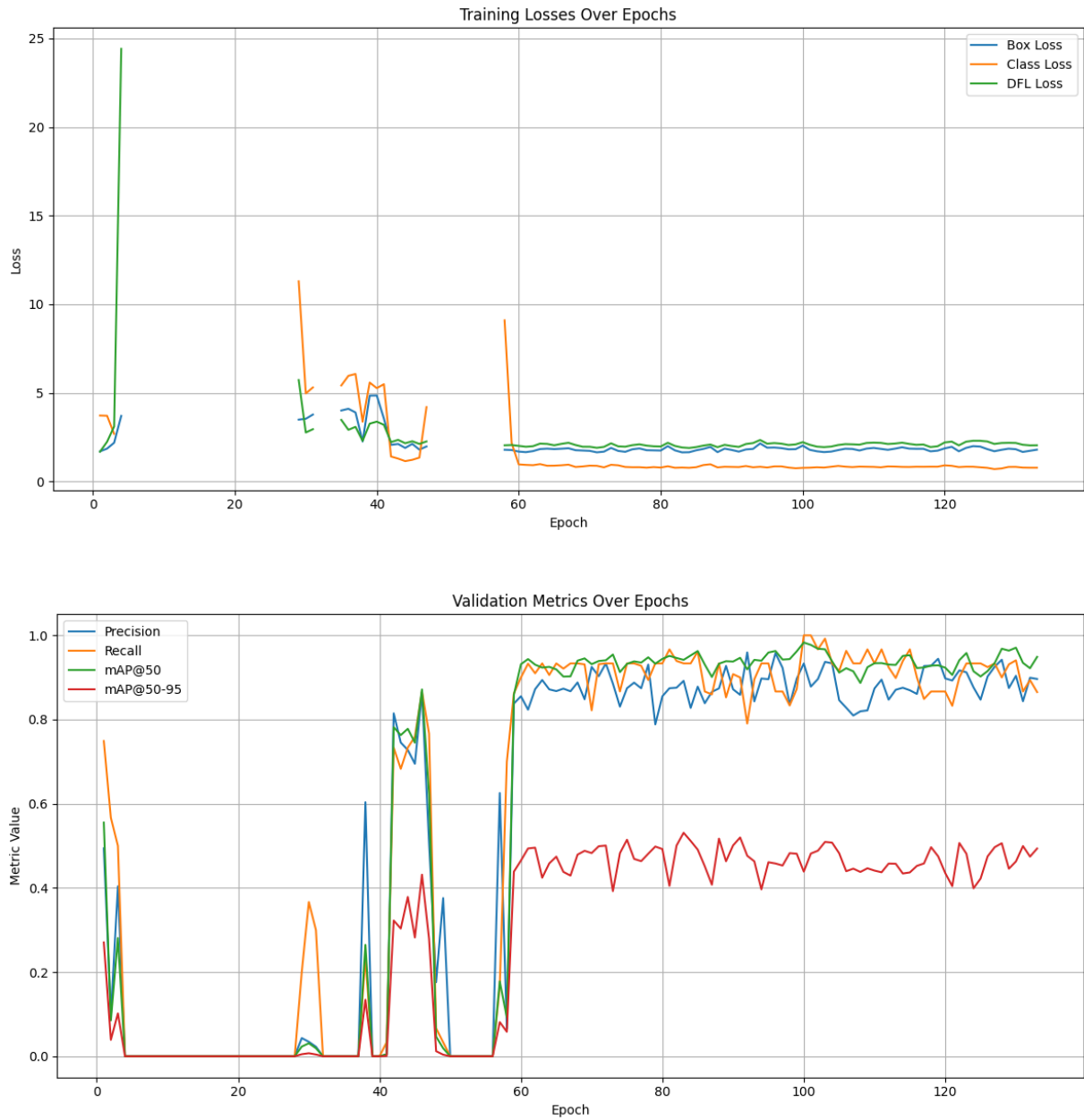


Figure 4. YOLOv11x training and validation performance

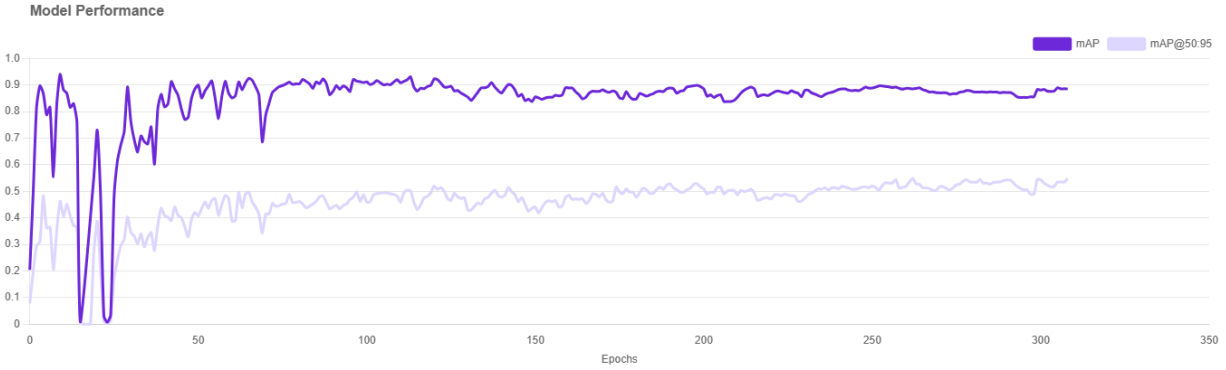


Figure 5. Model performance curves in RoboFlow

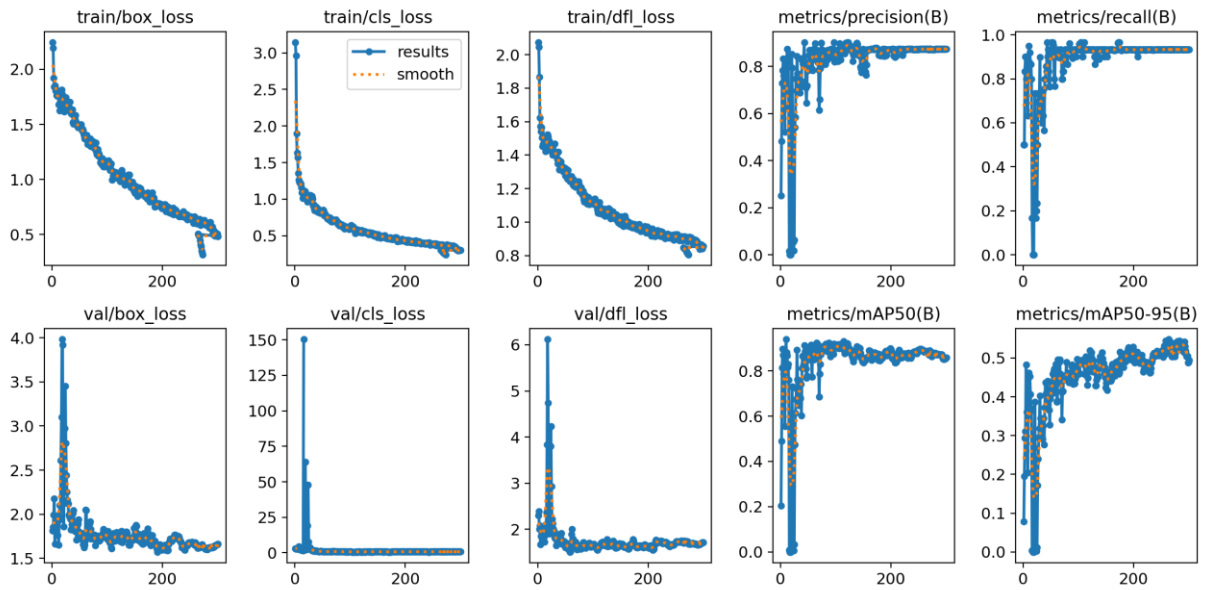


Figure 6. Results in RoboFlow

The right panel presents the validation metrics over epochs:

Precision – proportion of correct detections among predicted positives

Recall – proportion of true objects correctly detected

mAP@50 and **mAP@50–95** mean Average Precision at IoU thresholds 0.5 and 0.5–0.95.



Figure 7. Prediction results for Yolo11n



Figure 8. Prediction results for Yolo11x



Figure 9. Prediction results for Yolo11 Accurate in Roboflow

5. Discussion and Comparative Analysis

The comparative evaluation of the three YOLOv11 variants—YOLOv11n (Nano), YOLOv11x (Extreme), and YOLOv11 (Accurate)—demonstrated clear distinctions in model performance, convergence behavior, and deployment suitability. Across all experiments, YOLOv11n consistently achieved a balanced trade-off between detection accuracy and computational efficiency, whereas YOLOv11x and YOLOv11 (Accurate) showed higher capacity but at the cost of increased training complexity and inference time.

During the Colab-based training sessions, the YOLOv11n model exhibited stable convergence, with all loss components (box, class, and distribution focal loss) decreasing gradually and validation metrics stabilizing after approximately 83 epochs. The model achieved a precision of 0.747, recall of 0.831, and mAP@50 of 0.781, outperforming the larger YOLOv11x model, which yielded 0.641 precision, 0.809 recall, and mAP@50 of 0.758. The smaller YOLOv11n model demonstrated less overfitting and more consistent generalization, making it preferable for lightweight, edge-based agricultural applications such as real-time strawberry ripeness monitoring in greenhouse environments.

In contrast, the YOLOv11x model displayed greater fluctuations in validation metrics, indicating higher sensitivity to data imbalance and potential overfitting. Although the total loss converged, the validation mAP values did not improve substantially beyond those of YOLOv11n. This suggests that the larger parameter count and deeper architecture of YOLOv11x were not fully exploited under the available dataset size and training regime.

The YOLOv11 (Accurate) model trained on the Roboflow workspace achieved the best overall accuracy, with an mAP@50 of 88% on the validation set and 81% on the test set. These results were supported by smooth loss curves and stable model-performance trends, where both object and box loss steadily decreased throughout 300 training epochs. The high precision and recall values (both approximately 0.88) indicate strong confidence calibration and reliable bounding-box localization. This performance can be attributed to longer training duration, improved data augmentations, and optimized hyperparameter scheduling automatically applied within the Roboflow environment.

Overall, these findings highlight that model selection should depend on application constraints. YOLOv11 (Accurate) is ideal when maximum detection precision is required and computational resources are available. YOLOv11n, however, remains more suitable for real-time edge deployment on embedded devices such as the NVIDIA Jetson platform, offering an optimal balance between accuracy and inference speed. Future work should explore hybrid training strategies that combine the robustness of the Roboflow pipeline with the lightweight adaptability of Colab-based fine-tuning for on-device performance optimization.

6. Conclusion

This study evaluated three versions of the YOLOv11 architecture—YOLOv11n, YOLOv11x, and YOLOv11 (Accurate)—for automated strawberry ripeness detection. The experimental results demonstrated that while all three models were capable of accurately identifying ripe fruits, their performances varied significantly in terms of precision, recall, and computational efficiency.

The YOLOv11n model, trained in Google Colab, provided the most balanced results, achieving 0.78 mAP@50 with stable convergence and low computational overhead, making it well-suited for real-time edge deployment on devices such as NVIDIA Jetson. The YOLOv11x model achieved comparable recall but showed signs of overfitting and less stable metric trends, indicating that its higher capacity was underutilized with the available dataset size.

The YOLOv11 (Accurate) model trained through Roboflow achieved the highest accuracy (mAP@50 = 88%) on the validation set and 81% on the test set, demonstrating the benefits of extended training epochs and advanced augmentation pipelines. Its superior precision and recall confirm its potential for high-accuracy offline or cloud-based agricultural analysis, though at a higher computational cost.

In conclusion, YOLOv11n is recommended for real-time and edge-based monitoring systems, whereas YOLOv11 (Accurate) is preferable for precision-focused research or production-scale deployments requiring maximum detection reliability. Future research should explore knowledge distillation or hybrid fine-tuning techniques to combine the accuracy of larger models with the speed and efficiency of lightweight versions for practical, field-ready precision agriculture applications.